

# Tao Ma

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| CONTACT INFORMATION        |  <a href="https://xiaobaobaochifan.github.io">xiaobaobaochifan.github.io</a><br> <a href="mailto:tao.ma@ucl.ac.uk">tao.ma@ucl.ac.uk</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  <a href="https://linkedin.com/in/tao-ma-93642013a">linkedin.com/in/tao-ma-93642013a</a><br> <a href="https://github.com/xiaobaobaochifan">github.com/xiaobaobaochifan</a> |
| ACADEMIC POSITION          | <b>University College London (UCL)</b><br>Global Business School for Health<br>Lecturer (Assistant Professor) in Health Informatics and Statistics<br><br><b>London School of Economics and Political Science (LSE)</b><br>Research Assistant<br>Mentor: Prof. Qiwei Yao                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Apr 2026 – Present<br>London<br><br>Oct 2025 – Dec 2025<br>London                                                                                                                                                                                                                                                                            |
| EDUCATION                  | <b>London School of Economics and Political Science (LSE)</b><br>Doctor of Philosophy in Statistics<br>Advisor: Prof. Zoltán Szabó<br><br><b>New York University (NYU)   Courant Institute</b><br>Master of Science in Mathematics<br><br><b>Hong Kong Baptist University (HKBU)</b><br>Bachelor of Science in Statistics and Operations Research (Science Elite Program)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Sept 2021 – Sept 2025<br>London<br><br>Sept 2016 – Aug 2018<br>New York<br><br>Sept 2012 – Aug 2016<br>Hong Kong                                                                                                                                                                                                                             |
| RESEARCH INTERESTS         | <ul style="list-style-type: none"><li>• Theory &amp; methodology: reinforcement learning, large language models, deep generative models.</li><li>• Trustworthy &amp; interpretable AI: sustainability, sparse autoencoders, offline learning, causality.</li><li>• Applications: healthcare, supply chains, organizational change, time-series forecasting, robotics.</li><li>• AI behaviours: persuasion, sycophancy, deliberation, policy, social risks.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                              |
| PUBLICATIONS AND PREPRINTS | <p><b>Ma, T.</b>, Zhu, J., Cai, H., Qi, Z., Chen, Y., Shi, C., &amp; Laber, E. B. (2026). Sequential Knockoffs for Variable Selection in Reinforcement Learning. <b>Journal of the American Statistical Association (JASA)</b>. </p> <p>Schoenegger, P., ..., <b>Ma, T.</b>, et al. (2025+). When Large Language Models are More Persuasive Than Incentivized Humans, and Why.  </p> <p>Ghilardi, D., Belotti, F., Molinari, M., <b>Ma, T.</b>, &amp; Palmonari, M. (2025). Group-SAE: Efficient Training of Sparse Autoencoders for Large Language Models via Layer Groups. <b>EMNLP</b>. </p> <p><b>Ma, T.</b>, Yang, X., &amp; Szabó, Z. (2025+). To Switch or not to Switch? Balanced Policy Switching in Offline Reinforcement Learning. </p> <p><b>Ma, T.</b>, &amp; Yang, X. (2024). A Framework for Policy Evaluation Enhancement by Diffusion Models. Tiny Papers <b>ICLR</b>. (<b>Invite to present</b>)</p> |                                                                                                                                                                                                                                                                                                                                              |
| SELECTED ACHIEVEMENTS      | <ul style="list-style-type: none"><li>• LSE-NYU Research Seed Fund 2025 – 2026</li><li>• LSE PhD Studentship 2021 – 2025</li><li>• LSE Class Teacher Award 2022 – 2023</li><li>• Best TA Teaching Award (HKUST) 2019 – 2020</li><li>• HKSAR Government Scholarship (highest honor for undergraduates in Hong Kong) 2013 – 2016</li><li>• Outstanding Student Award (7/4134, by The Baptist Convention of Hong Kong) 2015 – 2016</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                              |

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|                        | <ul style="list-style-type: none"> <li>• Scholastic Award 2015 – 2016</li> <li>• APEC Scholarship (by Asia Pacific Economic Cooperation) 2014 – 2016</li> <li>• Best Debater prize at The World Mandarin Debating Championship, Hong Kong 2015 – 2016</li> <li>• Summer Undergraduate Research Fellowship 2014 – 2015</li> <li>• Mr. Li Men Jan Prize in Mathematics 2014 – 2015</li> <li>• Department of Mathematics Outstanding Academic Performance Award 2013 – 2014</li> <li>• Sir Tseng Chi Lu Scholarship 2013 – 2014</li> <li>• President's Honour Roll 2012 – 2016</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| SKILLS                 | Python, R, SAS, MATLAB, $\text{\LaTeX}$ , HTML, CSS, JavaScript, LEAN, Bash, PyTorch, VS Code, Linux, Git, Amazon Web Services, Google Cloud Platform, local HPC management                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| TEACHING<br>EXPERIENCE | <p><b>University College London</b> London</p> <p><i>Lecturer/Module Co-Lead</i></p> <ul style="list-style-type: none"> <li>• <b>Undergraduate courses:</b> <ul style="list-style-type: none"> <li>◇ GBSH0050 Introduction to Statistics and Data Insights Jan 2027 – Mar 2027</li> <li>Introductory statistics and machine learning</li> </ul> </li> <li>• <b>Graduate courses:</b> <ul style="list-style-type: none"> <li>◇ GBSH0009 Research Project Apr 2026 – Present</li> <li>Research project course for Master's students</li> <li>◇ CHME0016 Artificial Intelligence in Healthcare Jan 2027 – Mar 2027</li> <li>Machine learning algorithms and AI methods for healthcare with Python</li> </ul> </li> </ul> <p><b>London School of Economics and Political Science</b> London</p> <p><i>Class Teacher/Teaching Assistant</i></p> <ul style="list-style-type: none"> <li>• <b>Undergraduate courses:</b> <ul style="list-style-type: none"> <li>◇ ST101 Programming for Data Science (Python) Sept 2022 – Dec 2024</li> <li>NumPy, Pandas, testing, object-oriented programming, functional programming</li> <li>◇ ST107 Quantitative Methods Jan 2022 – May 2022</li> <li>Introductory statistics for students in social science</li> <li>◇ ST102/109/110 Elementary Statistical Theory Sept 2022 – May 2025</li> <li>One-year course in probability and statistics</li> </ul> </li> <li>• <b>Graduate courses:</b> <ul style="list-style-type: none"> <li>◇ ST445 Managing &amp; Visualizing Data (Python) Sept 2021 – Dec 2021</li> <li>Pandas, BeautifulSoup, Sklearn, Jupyter Notebook</li> </ul> </li> </ul> <p><b>Hong Kong University of Science and Technology (HKUST)</b> Hong Kong</p> <p><i>Research Assistant</i> Oct 2018 – Nov 2019</p> <ul style="list-style-type: none"> <li>◇ Mathematical physics, partial differential equations, signal processing</li> </ul> <p><i>Class Teacher</i> Feb 2020 – May 2020</p> <ul style="list-style-type: none"> <li>◇ MATH 2011 Introduction to Multivariable Calculus</li> </ul> <p><b>New York University</b> New York</p> <p><i>Recitation Leader</i> Sept 2017 – Dec 2017</p> <ul style="list-style-type: none"> <li>◇ MATH-UA 211 Mathematics for Economics</li> </ul> |

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| ACADEMIC SERVICE        | <b>Technology Innovation and Data-Informed Healthcare Symposium</b>                                                                                                                                                             | Sept 2026 – Oct 2026   |
|                         | <i>Reviewer Panel</i>                                                                                                                                                                                                           |                        |
|                         | <b>International Conference on Econometrics and Statistics (EcoSta 2026)</b>                                                                                                                                                    | Aug 2026               |
|                         | <i>Session Chair</i>                                                                                                                                                                                                            |                        |
|                         | <b>LSE Statistics Research Showcase</b>                                                                                                                                                                                         | Apr 2025               |
|                         | <i>Session Chair</i>                                                                                                                                                                                                            |                        |
|                         | <b>Neural Information Processing Systems (NeurIPS)</b>                                                                                                                                                                          | May – Sept 2024 – 2026 |
|                         | <i>Reviewer</i>                                                                                                                                                                                                                 |                        |
|                         | <b>International Conference on Machine Learning (ICML)</b>                                                                                                                                                                      | Feb 2025 – May 2025    |
|                         | <i>Reviewer</i>                                                                                                                                                                                                                 |                        |
|                         | <b>International Conference on AI and Statistics (AISTATS)</b>                                                                                                                                                                  | Oct – Jan 2024 – 2026  |
|                         | <i>Reviewer</i>                                                                                                                                                                                                                 |                        |
|                         | <b>International Conference on Learning Representations (ICLR)</b>                                                                                                                                                              | Oct – Jan 2024 – 2027  |
|                         | <i>Reviewer</i>                                                                                                                                                                                                                 |                        |
|                         | <b>London School of Economics and Political Science</b>                                                                                                                                                                         | London                 |
|                         | <i>Committee member</i>                                                                                                                                                                                                         |                        |
|                         | • Department Review Representative                                                                                                                                                                                              | 2024                   |
|                         | • Department Teaching Committee                                                                                                                                                                                                 | 2023 – 2025            |
|                         | • Department Moving Plan Committee                                                                                                                                                                                              | 2023 – 2025            |
|                         | <b>IMS Annual Meeting</b>                                                                                                                                                                                                       | June 2022              |
| PROFESSIONAL EXPERIENCE | <i>Sessions Coordinator</i>                                                                                                                                                                                                     | London                 |
|                         | <b>University Mandarin Debate Team</b>                                                                                                                                                                                          | Sept 2012 – Aug 2016   |
|                         | <i>President, coach</i>                                                                                                                                                                                                         | Hong Kong              |
|                         | • Organized weekly training sessions, managed administrative tasks, participated in international competitions and gave 3-round interviews to around 100 applicants each year.                                                  |                        |
|                         | <b>Professional Swimming</b>                                                                                                                                                                                                    | 2004 – 2009            |
|                         | Individual Medley (200 meters), Breaststroke (100 meters), Butterfly Stroke (100 meters)                                                                                                                                        | Beijing                |
|                         | <b>Invited Talks</b>                                                                                                                                                                                                            |                        |
|                         | Balanced Policy Switching in Reinforcement Learning.<br>The 14th IASC-ARS & SoAI 2026 Joint Conference, Chiang Mai, Thailand.                                                                                                   | Dec 2026               |
|                         | Balanced Policy Switching in Reinforcement Learning.<br>The 9th International Conference on Econometrics and Statistics (EcoSta 2026),<br>Kyoto, Japan.                                                                         | Aug 2026               |
|                         | Balanced Policy Switching in Reinforcement Learning.<br>Workshop at Representing, calibrating & leveraging prediction uncertainty from<br>statistics to machine learning, Isaac Newton Institute & University of Cambridge, UK. | May 2025               |
| INVITED TALKS           | Balanced Policy Switching in Reinforcement Learning.<br>Statistics Research Showcase, LSE, UK.                                                                                                                                  | Apr 2025               |
|                         | Careers in Data Science and AI.<br>The Alan Turing Institute & The Brilliant Club, UK.                                                                                                                                          | Feb 2024               |
|                         | Sequential Knockoffs for Variable Selection in Reinforcement Learning.<br>PhD Presentation Events, LSE, UK.                                                                                                                     | May 2023               |
|                         | Sequential Knockoffs for Variable Selection in Reinforcement Learning.<br>Prof. Tengyao Wang's Reading Group, LSE, UK.                                                                                                          | Apr 2023               |
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|           | Sequential Knockoffs for Variable Selection in Reinforcement Learning.<br>Prof. Qiwei Yao's Reading Group, LSE, UK.                                                                                                                                                                                    | Mar 2023            |
|           | Sequential Knockoffs for Variable Selection in Reinforcement Learning.<br>RL+X Online Seminar.                                                                                                                                                                                                         | Feb 2023            |
|           | Reinforcement Learning with Representation Learning.<br>LSE PhD Reading Group, UK.                                                                                                                                                                                                                     | Oct 2022            |
|           | Nonstationarity in a Markov Decision Process.<br>RL+X Online Seminar.                                                                                                                                                                                                                                  | June 2022           |
|           | Sequential Knockoffs for Variable Selection in Reinforcement Learning.<br>Statistics Research Showcase (Poster Section), LSE, UK.                                                                                                                                                                      | June 2022           |
| WORKSHOPS | A personal literature wiki, written and maintained by an AI agent.<br>FORRT AI in Metascience Conference.  <br>With Daryl Y. H. Lee. | Sept 2026<br>Online |
| LANGUAGES | Mandarin (native), English (fluent), Cantonese (fluent)                                                                                                                                                                                                                                                |                     |
| GAMES     | Poker, Mahjong, Pokémon, Mario, Kirby, Dragon Ball, NBA 2K, Asphalt, FIFA, Horizon 4, Zelda, GTA 5/6 and a lot of others                                                                                                                                                                               |                     |